USF2000 Aero Data Analysis

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1 Introduction

In this document, three aero setups are compared in MoTeC i2 to assess their performance. These stints were completed during an FP1 session, and as shown in Table 1, there is an out-lap and two-timed laps per run. The comparison between the performance gains or losses of each setup will use their fastest lap unless additional detail can be obtained with the data from the other two laps. For reference during the discussion of the results, the track layout is given in Figure 1.

Table 1: Aero setups time summary.

Run Romeo	Run Sierra	Run Tango
Out-lap: 2:02.997 s	Out-lap: 2:03.142 s	Out-lap: 2:01.274 s
Lap 1: 1:43.227 s	Lap 1: 1:42.746 s	Lap 1: 1:42.031 s
Lap 2: 1:42.841 s	Lap 2: 1:42.363 s	Lap 2 1:42.098 s

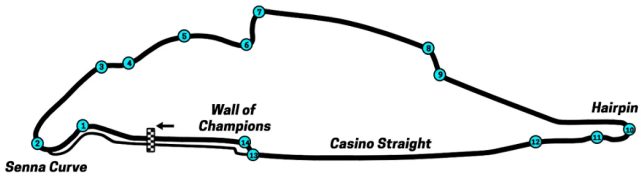

Figure 1: Circuit Gilles Villeneuve layout.

2 Drag Analysis

2.1 Braking and speed development

The losses in lap time induced by drag become increasingly relevant as the car reaches higher speeds. Some examples of these high-speed sectors marked in Figure 2 are:

- The Casino Straight (region **D**)
- The straight between the Wall of Champions and turn 1 (region **E**)
- The exit of turn 9 up to the Hairpin of turn 10 (region **C**)

Figure 2 also shows the lap time variance, which is the cumulative time difference between the selected laps per channel. The variance is generated as a curve with either downward or upward slope regions showing where time is gained or lost respectively. As Run Tango (RT) is the fastest lap of the session, it is overlaid upon Run Romeo (RR) and Run Sierra (RS).

Additional detail on these regions can be identified at the braking points where there are momentary peaks (upward slopes), indicating that RT is losing time. RT which is seen to reduce lap time on the straights running this low-drag configuration loses on downforce potential and appears to not carry as much speed into the corner.

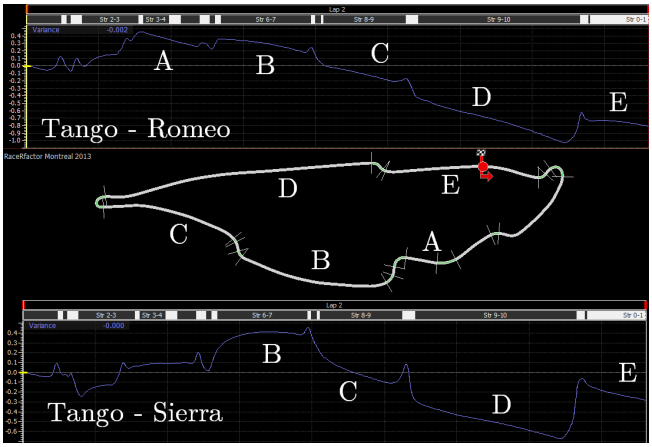

Figure 2: Time variance comparison between the slower laps with RT overlaid.

To visualise this, Figure 3 shows a comparison between the vehicle speed, brake throttle application, and longitudinal acceleration channels of the three analysed laps in distance mode. The following points are summarised to show the influence of drag related effects in drivability:

- The top end speed of RT is noticeable higher than that of RR and RS at the location of the cursor at 3683 metres. The second highest speed is achieved by RS with a decrease of 6.4 km/h as shown in the channel list marked in red. This order is repeated at regions A and B, with RT being the fastest, then RS and lastly RR with the lowest top end speed
- Region C showcases the effects of drag on acceleration. Although RT appears to not have an optimum exit out of T7, the rate of change in speed is sufficient to not only match the speed of the other two setups, but to surpass them by an average of 4 km/h at the end of the straight
- Region D and similar instances around the lap show that RT had to brake earlier. At the cursor, RT is applying 22% of the brakes, whilst RS has barely started braking with 1.3% of pedal position and RR is able to brake later
- By assessing the similarity in magnitude of longitudinal accelerations, it can be concluded that this is not a driving technique and that it is related to the downforce capabilities of the car. More detail on downforce will be given in the next section

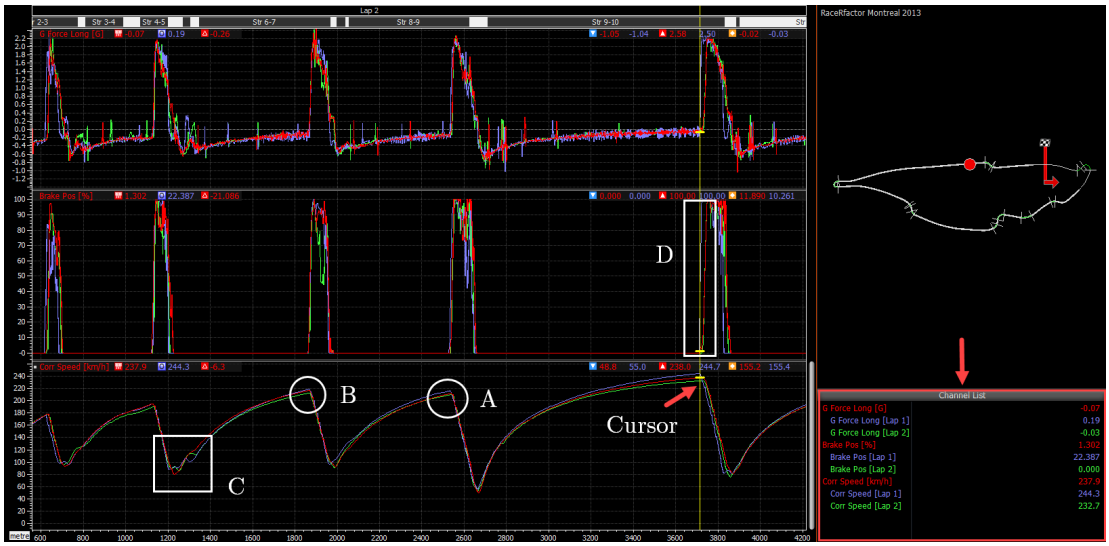

Figure 3: Telemetry channel comparison between RT (blue), RS(red) and RR(green).

3 Downforce

When assessing the downforce of a car, it is important to define a set of reference variables that directly affect the development of aerodynamic load. Such parameters used in this study are ride heights (FRH & RRH), peak lateral and longitudinal accelerations (α_x & α_y), and suspension forces (FSF & RSF). Based on the trends of these data sets at specific track conditions, the aero balance and front and rear loads can be estimated.

3.1 Total downforce

After investigating the $FRHs$ and $RRHs$ of the three setups, the total downforce can be estimated based on previous comments about induced drag. Figure 4 shows an initial comparison between RHs (regions **1** & **2**) under steady state high-speed conditions ($\alpha_x \approx 0$ & $\alpha_y \approx 0$). From these channels, and their values indicated in the list marked in red, it is evident that:

- RR is the one running closest to the ground, maintaining this trend throughout the lap and potentially generating the most downforce (e.g. suspension with more load)
- RS's RHs are a mid-configuration between RR and RT. In particular, its FRH (region 1) is consistently closer to that of RR by ≈ 1 mm of difference, whilst the largest difference is identified at the RRH (region 2) often showing variations of ≈ 4 mm
- RT has the highest RHs which can be observed in the scatterplot on the top left of Figure 4. The arc drawn by the traces represents the upcoming braking zone at the end of the Casino Straight

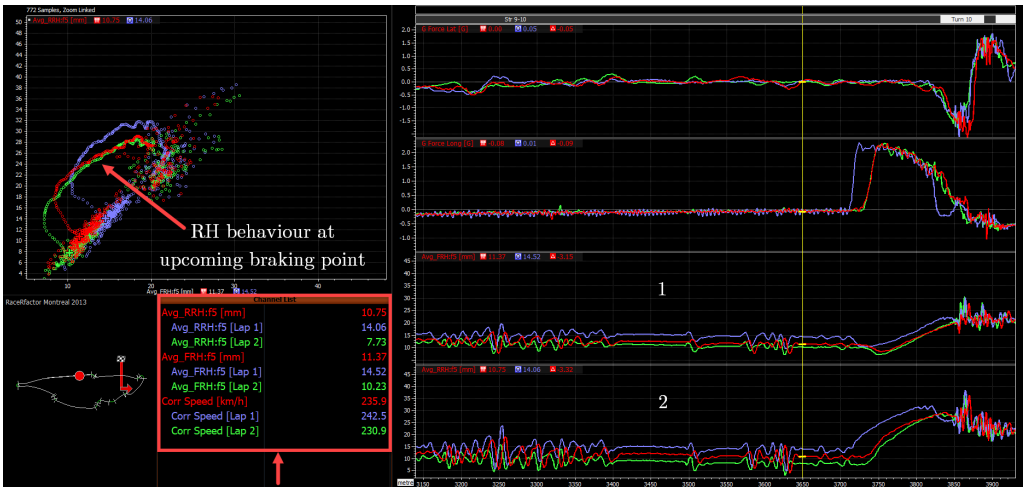

Figure 4: RH comparison between setups at the Casino Straight and upcoming braking point.

Recalling the negative slopes in Figure 2, these results can explain how RT was reducing lap time in this particular circuit with relatively long straights. Respectively, in Figure 5 the total suspension forces preemptively deem RR as the setup with the most downforce, then RS and lastly RT.

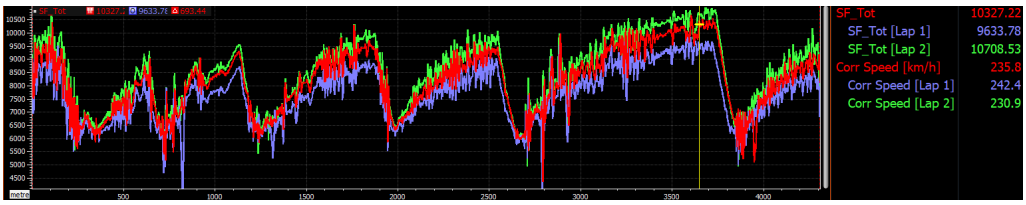

Figure 5: Total average suspension force channel defined as: $SF_{TOT_{avg}} = FSF_{avg} + RSF_{avg}$.

3.2 Aero balance: front and rear downforce interaction

To understand the contribution of aerodynamic loads at each axle, it is useful to explore the initial conditions of each setup on their respective out laps. The FRH and RRH channels show similar attitudes in the setups with the presence of ≈ 4 mm of rake. More specifically, FRH @ $RR = 23.61\ mm$, @ $RS = 22.37\ mm$ and @ $RT = 22.95\ mm$. Therefore, the difference in suspension forces between setups is a cause of the aero balance (AB) and overall downforce of the car. $FSFs$ and $RSFs$ are presented in Figure 6, where RR appears to have less AB (rear-biased) than RS.

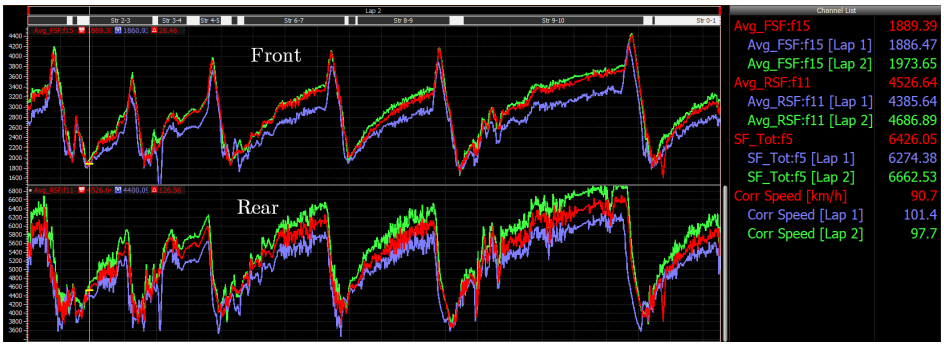

Figure 6: Based on the similarity between $FRHs$ and $FSFs$ of RR and RS, it is now clear that the former has more rear aero load as its RRH is lower specially during high-speed sectors. It is expected to have an $AB > 50\%$.

By comparing the G-G diagrams of RS and RT, the aero balance of the fastest setup can be properly assessed. The accelerations out of a corner are important for this comparison; therefore, Figure 7 only shows the setups with high data density on the binned G-G diagrams (second row). Firstly, RT has a more consistent braking as demonstrated by the tightly packed region at $(-a_y)$ and $a_x = 0$ values, although it does not carry as much speed into the corner as previously seen. The higher point population of RT at more negative a_y and a_x values indicates that this low-downforce setup has an acceptable level of rear support when coming out of a corner. Therefore, it can be concluded that its $AB < 50\%$, similar to RR.

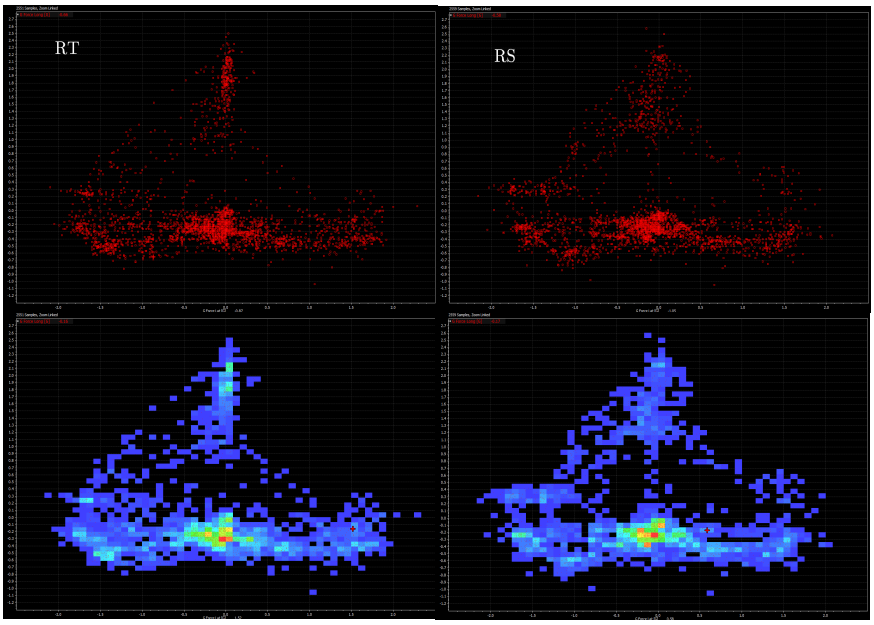

Figure 7: G-G diagram comparison between RT and RS.